

Edge-AI Airspace Surveillance System

A distributed IoT early warning system combining edge AI vision, distance and speed tracking, and remote MQTT alerting.

Project summary:

In this project we will be using Raspberry Pi as the brain of the system, which uses the Pi camera module to receive a video stream and uses the YOLOv8 model to classify safe and hazardous movement in the airspace by detecting if it is a commercial plane, civil helicopter, or own military airborne/drone, which are safe, and detecting if it is a missile, airborne object, or drone from another country, which is hazardous. Once it detects the movement, it calculates the distance and speed. The system uses an Arduino Uno WiFi Rev with an ultrasonic/LiDAR sensor and publishes the data via MQTT. The type of object detected, like a helicopter, drone, plane, or missile; its distance; speed; and whether it is safe or not are shown on an MQTT-based dashboard using Node-RED. Once a hazardous movement is detected, the ESP32 receives MQTT threat alert from a different location for triggering the alarm

Node 1: A Raspberry Pi with a Pi camera runs YOLO object detection and hosts the MQTT dashboard (Node-RED).

Node 2: Arduino UNO WiFi Rev2 uses the ultrasonic sensor (Lidar if available) to calculate target speed/distance and publishes it via MQTT.

Node 3: ESP32 placed in a different location receives MQTT threat alerts from the Pi to trigger local alarms.

BOM: Raspberry Pi, Pi Camera, Arduino UNO WiFi Rev2, ESP32, Ultrasonic Sensor (Lidar if available), and Active Piezo Buzzer.

